



YISHUN INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAM
Higher 2

NAME

INDEX NO

CG

BIOLOGY

9744/03

Paper 3 Long Structured and Free-Response Questions

13 Sep 2022

Candidates answer on the Question Paper.

2 hours

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name and class in the spaces at the top of this page.

Write in dark blue or black pen on both sides of the paper.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

Section A

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **one** question in the spaces provided on the Question Paper.

Indicate the question you have attempted at the top of page **18**.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	30
2	10
3	10
Section B	
4 or 5	25
Total	75

This document consists of **25** printed pages and **1** blank page.

Section A

Answer **all** questions in this section.

1 Estrogen receptors (ER) are expressed in mammary cells. Fig. 1.1 shows how an ER serves its functions at target cells:

- as an intracellular receptor activated by the steroid hormone estrogen (E), when E binds to complementary ligand binding site on ER,
- and as a transcriptional activator, after two ER bound with E dimerised and enters the nucleus.

Estrogens induce transcription of genes in their target cells. Such genes have estrogen response elements (ERE) as enhancers.

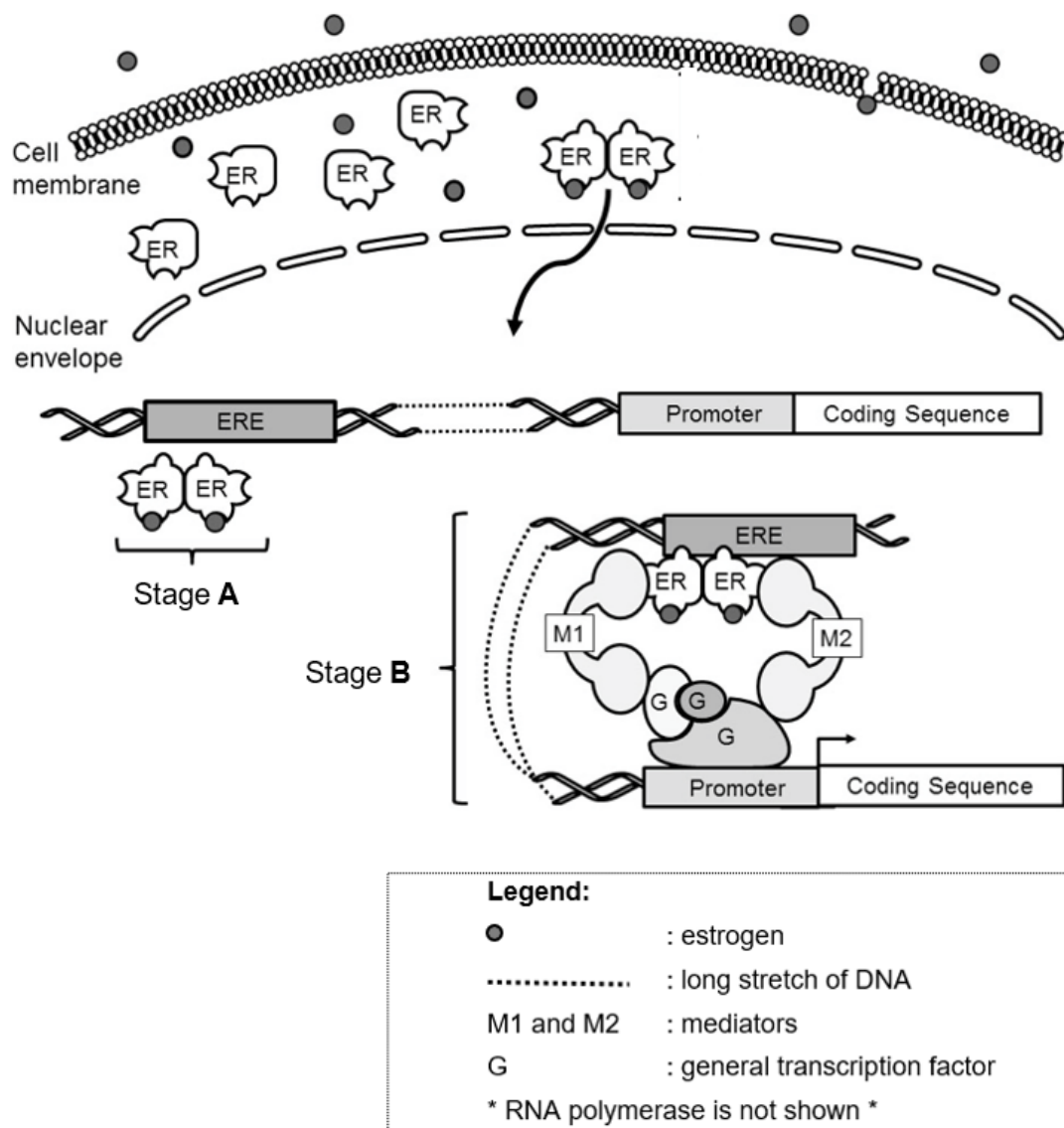


Fig. 1.1

- (a) With reference to Fig. 1.1, describe stages **A** to **B**.

Stage A:

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Stage B:

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[4]

Estrogen receptors are over-expressed in around 70% of breast cancer cases. Research into the treatment of metastatic breast cancer focused on the use of drugs such as selective estrogen-receptor modulator (SERM) and estrogen receptor downregulator (ERDR) as shown in Table 1.1.

Table 1.1

drug	function of drug	presence of drug-receptor complex in the nucleus	effect of drug on the target gene
SERM ○		✓	
ERDR ★		✗	Basal rate of transcription

(b) With reference to Table 1.1,

(i) explain how SERMs can be used in the treatment of breast cancer.

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- (ii) suggest why ERDRs might be more effective than SERMs.

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- (c) Cancer is described as a multi-step process. Explain how mutations may lead to uncontrolled proliferation in cells lining the milk ducts, leading to breast cancer.

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Fig. 1.2 shows that in breast cancer cells, there is an overexpression of human epidermal growth factor receptor 2 (HER2) proteins. HER2 functions as a receptor protein on the cell surface membrane to send a signal to the nucleus for the cell to undergo cell division.

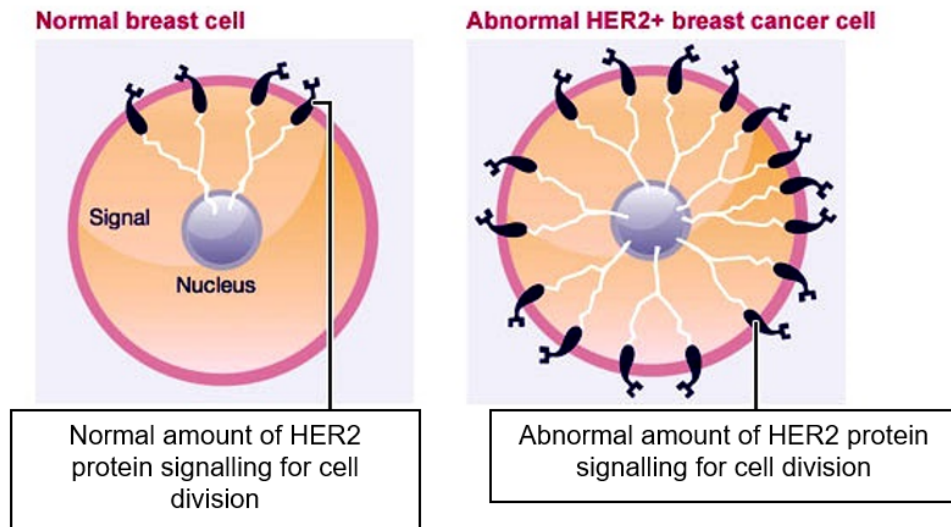


Fig. 1.2

Trastuzumab, an antibody specific to HER2, is an effective treatment for early-stage HER2+ breast cancer. However, advanced-stage HER2+ breast cancer patients tend to develop resistance to the treatment.

HER2+ breast cancer cells also overexpress CD47, a surface protein that helps cancer cells evade macrophage phagocytosis. Research has shown that blockage of CD47 using an antibody specific to CD47 (α -CD47 mAb) is also an effective treatment for HER2+ breast cancer.

An investigation was conducted to determine the effectiveness of treating HER2+ breast cancer cells with both trastuzumab and α -CD47 mAb.

Groups of genetically identical breast cancer cells were subjected to 3 different treatments:

1. incubation with α -CD47 mAb
2. incubation with trastuzumab
3. incubation with both trastuzumab and α -CD47 mAb

The investigation was done twice, with 3 replicates for each treatment.

The percentage of phagocytosis by macrophages was assessed and shown in Fig. 1.3.

Data are presented as mean $\pm$ S.D. (standard deviation).

Statistical differences between the two groups of cells are marked by asterisk ($P \leq 0.0001$).

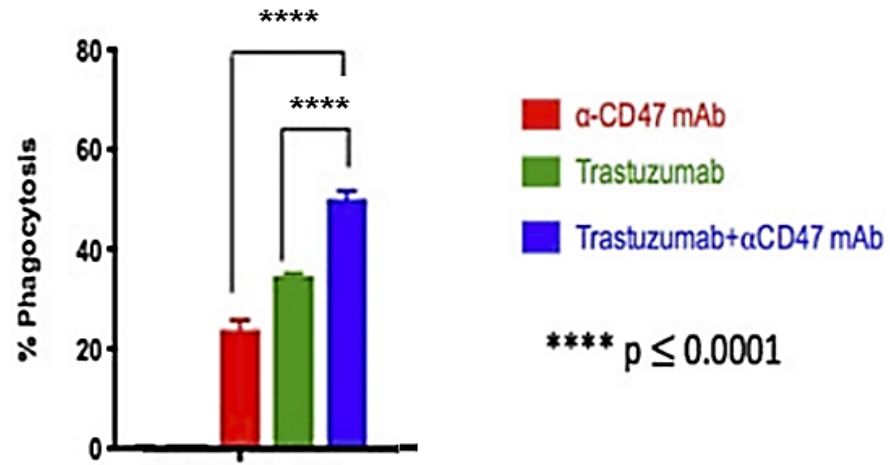


Fig. 1.3

- (d) Using Fig. 1.3 and the information provided, comment on the effectiveness of the combined treatment versus a treatment using either trastuzumab, or α -CD47 mAb alone in humans. Discuss on the confidence of the presented data.

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An early diagnosis of many types of cancer can result in successful treatment.

The BRCA2 protein is involved in suppressing the development of tumours.

Several different dominant alleles of the gene, *BRCA2*, code for faulty versions of the BRCA2 protein. The presence of any one of these faulty alleles leads to an increased chance of developing several types of cancer, including breast and lung cancers. Not everyone with one of these alleles develops cancer. This is because environmental factors, including lifestyle, are also involved.

Fig. 1.4 is a pedigree (family tree) showing the occurrence of cancers in four generations of a family. The presence of a faulty *BRCA2* allele was confirmed in person 15. The other individuals with cancer were not tested for the presence of the allele.

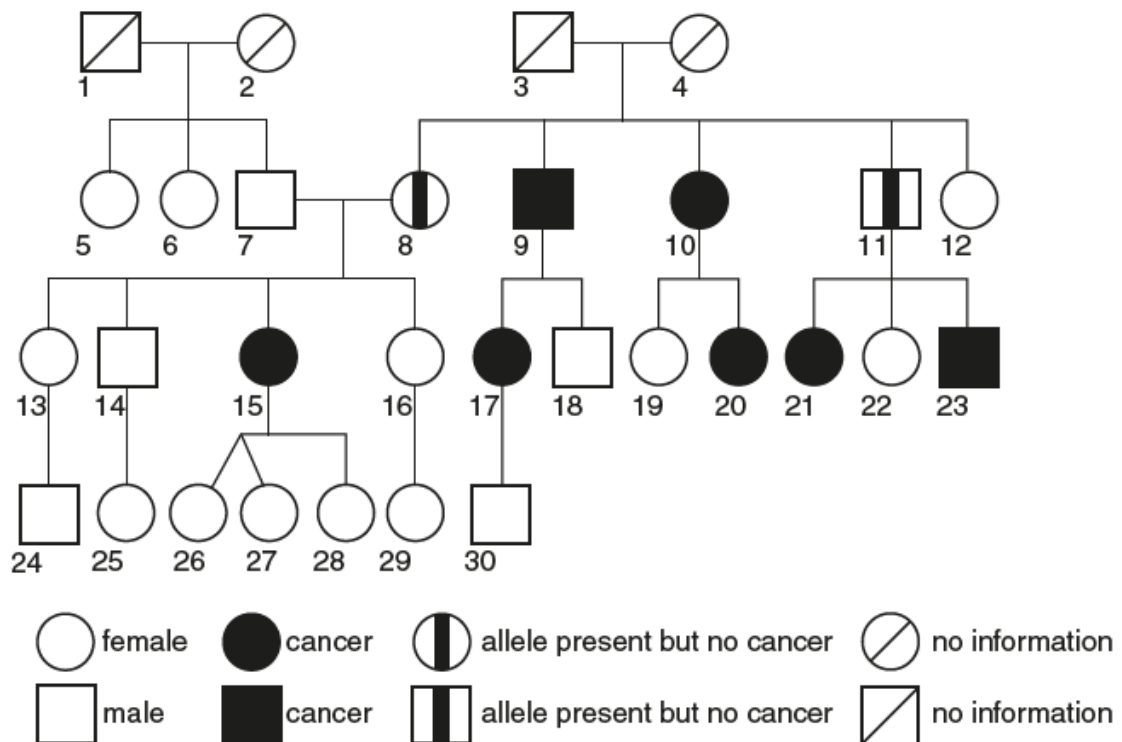


Fig. 1.4

- (e) Discuss the extent to which Fig. 1.4 provides evidence that a faulty *BRCA2* allele increases the risk of a person developing cancer.

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- (f) Traditionally, stem cells from bone marrow have been used to treat patients with leukaemia, a cancer of the blood or bone marrow.

Recent studies have shown that stem cells taken from umbilical cord blood may be more effective in treating leukaemia than stem cells taken from bone marrow.

Table 1.2 shows the results from patients with leukaemia after treatment with stem cells from unrelated sources over a two-year period.

Table 1.2

source of stem cells	probability / %	
	leukaemia-free survival	death from immune rejection of transplant
cord blood	34	12
bone marrow	33	31

Using information from Table 1.2 and your knowledge on stem cells, suggest why stem cells from the cord blood may be more preferred in the treatment of leukaemia.

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- (g) Embryonic stem cells have been known to express a specific set of genes that enable them to behave like stem cells, but these genes are turned off in normal somatic cells. Researchers have discovered that they can convert human fibroblast cells into embryonic stem cells by switching on the specific set of genes. These cells have been cultured in labs all over the world, and they are termed induced pluripotent stem cells (iPSCs).

- (i) Describe the characteristics of an induced pluripotent stem cell (iPSC).

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- (ii) Suggest an ethical concern regarding the use of embryonic stem cells which is no longer relevant with the development of iPSCs.

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The histone tails of different regions of chromatin are covalently modified during the formation of the human fibroblast cell. These modifications are added and removed by enzymes. The main effect of modifying histone tails is to attract specific proteins to a stretch of chromatin.

Fig. 1.5 shows the action of one of these enzymes, histone methyl transferase.

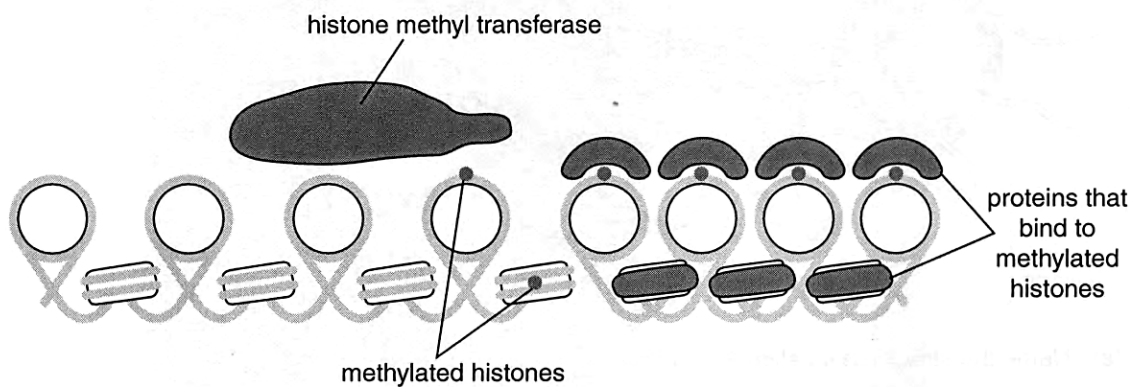


Fig. 1.5

- (iii) Suggest why histone methylation occurs over large areas of chromatin during the formation of the human fibroblast cell.

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[Total: 30]

- 2 Today, viruses are considered an exception to the cell theory which states that the simplest units of life are cells. Viruses have also been referred to as “organisms at the edge of life”.

(a) State **one** characteristic of viruses that may classify them as being

(i) living,

[1]

(ii) non-living.

[1]

Coronavirus disease 2019 (COVID-19) is an infectious disease caused by a novel coronavirus, severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2).

Fig. 2.1 shows the mechanism used by SARS-CoV-2 to infect the cells in the airways of human hosts.

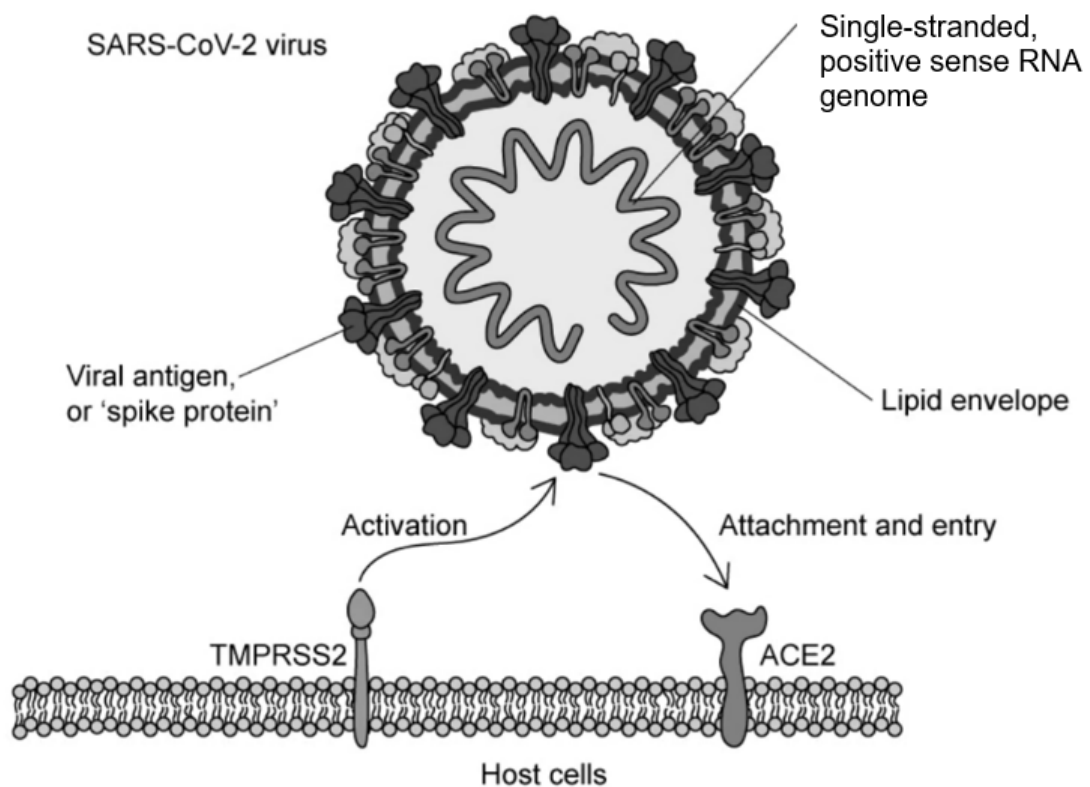


Fig. 2.1

- (b) With reference to Fig. 2.1, suggests how SARS-CoV-2 infects human cells.

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A drug currently being trialled for the treatment of COVID-19 is Nafamostat, a drug that is currently licensed for the treatment of other conditions.

During one trial, scientists investigated the effect of Nafamostat on the activity of TMPRSS2, and the results are shown in Fig. 2.2 below.

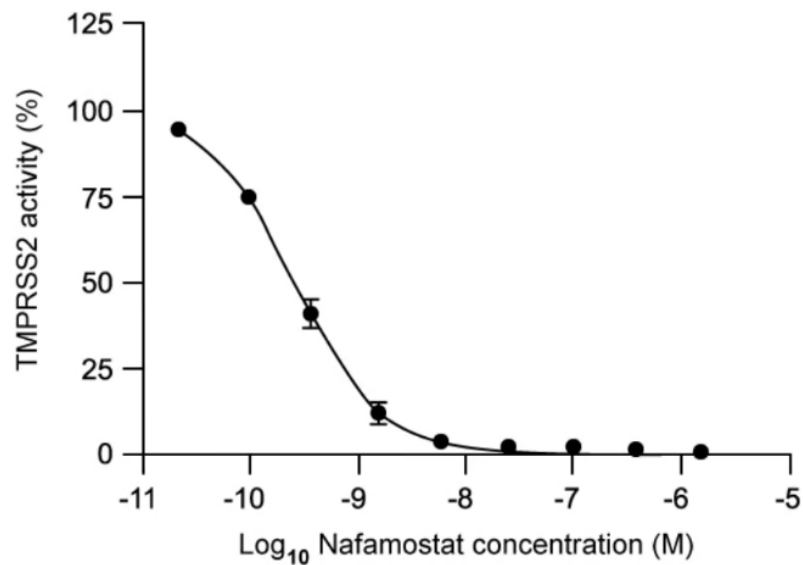


Fig. 2.2

- (c) With reference to Fig. 2.1 and and Fig. 2.2, suggest how Nafamostat might function as a treatment for COVID-19.

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Fig. 2.3 shows the reproductive cycle of the coronavirus SARS-CoV-2.

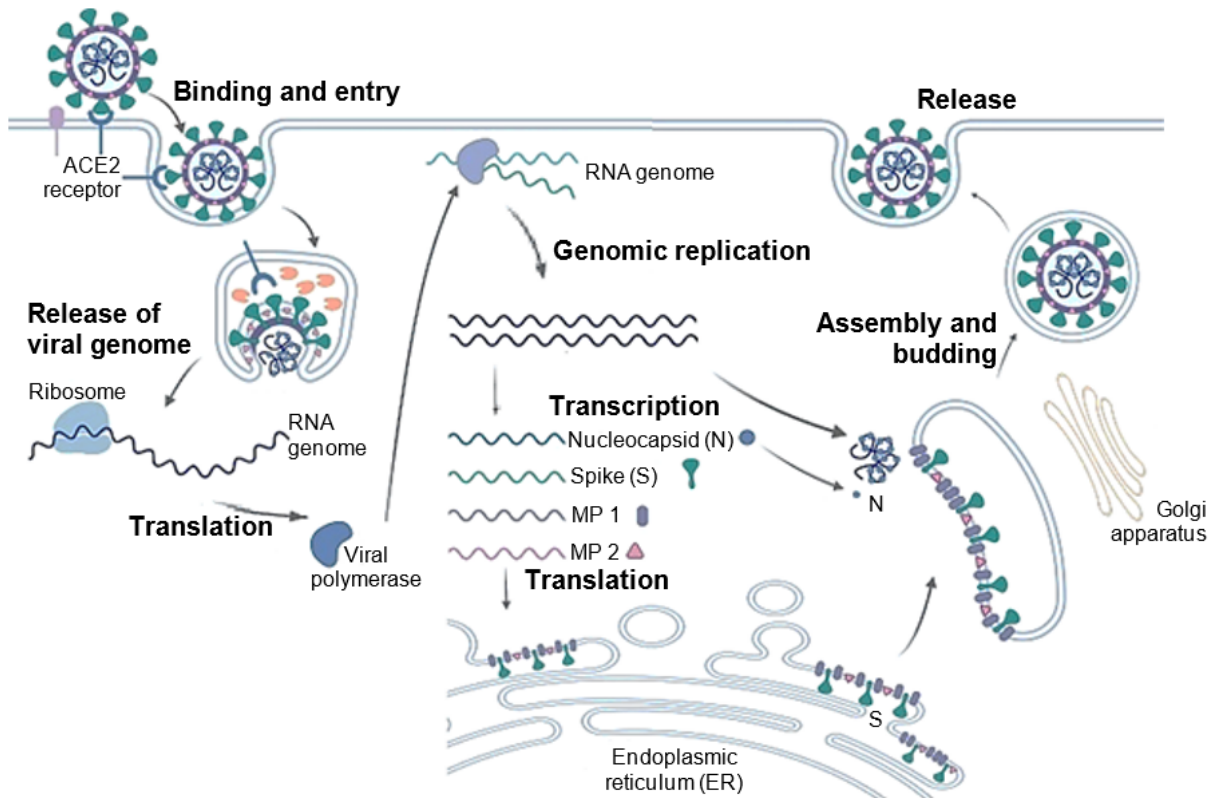


Fig. 2.3

(d) Compare between the reproductive cycles of SARS-CoV-2 and the influenza virus.

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[Total: 10]

- 3 Table 3.1 shows dengue incidences and precipitation pattern in a tropical country from 2013 to 2017.

Table 3.1

Year	Average annual rainfall/mm	Number of cases of Dengue Fever	Average Maximum Temperature/°C	Average Minimum Temperature /°C
2013	665	20270	32.3	23.2
2014	585	19902	33.9	23.2
2015	578	18910	33.8	24.2
2016	695	4260	34.8	24.5
2017	564	2145	36.2	25.0

- (a) With reference to Table. 3.1, comment on the relationship between dengue fever prevalence and climate factors.

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To reduce the transmission of dengue fever through decreasing vector numbers, scientists used high doses of radiation on male *A. aegypti* before releasing large numbers into the wild.

- (b) Suggest how releasing the treated male mosquitoes may reduce transmission of dengue fever.

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[2]

Recently a new method was developed to control *A. aegypti*. Scientists produced transgenic males carrying a 'lethal gene' which kills their offspring before they can reproduce.

The scientists released transgenic males every week in one area of a city in Brazil. At regular intervals they determined the number of *A. aegypti* per km² in the area where transgenic males were released and in a control area where no transgenic males were released.

Fig. 3.1 shows their results.

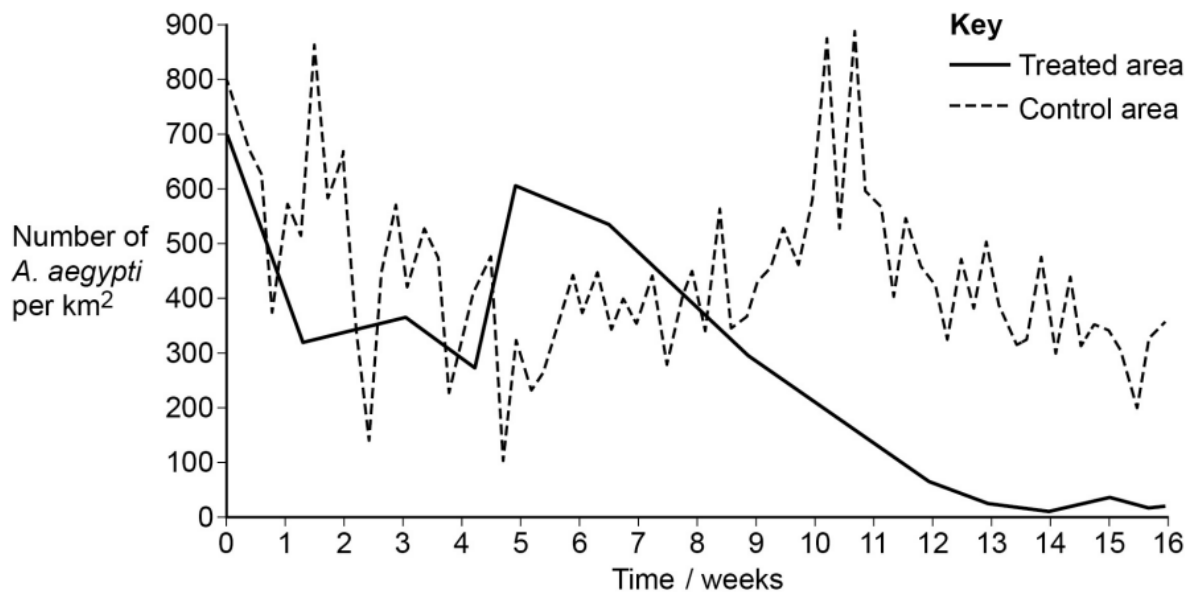


Fig. 3.1

- (c) (i) With reference to Fig. 3.1, conclude whether the release of transgenic males successfully reduced the vector population.

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- (ii) Suggest why the scientists released more transgenic males every week.

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[1]

[Total: 10]

Section B

Answer **ONE** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

You answers must be set out in parts **(a)**, **(b)**, etc., as indicated in the question.

- 4 (a)** DNA can be replicated *in vivo* (in cells) or *in vitro* (outside cells) by processes like the polymerase chain reaction (PCR).

Compare the process of DNA replication in cells with PCR. [10]

- (b)** Both DNA replication in cells and PCR may harbour mutations resulting in changes to DNA base sequences.

Discuss the impact of mutations to organisms. [15]

[Total: 25]

- 5 (a)** Both eukaryotic cells and bacterial cells are able to divide to give daughter cells. Eukaryotic cells can divide by mitosis to produce cells for growth and repair or meiosis to produce gametes. Bacterial cells, however, can only divide by binary fission to result in clones.

Compare the process of mitosis in eukaryotic cells and binary fission in bacterial cells. [10]

- (b)** Discuss how genetically varied cells can arise despite binary fission being the only mode of cell division for bacterial cells. [15]

[Total: 25]

Question

[illegible]

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